

or downloaded software. It enters a computer by hiding itself inside a legitimate program—often a game, a screen saver or a utility. It then puts malicious code into the operating system, which enables the attacker to gain access or control of the compromised computer.

Some Trojan horses may be used by the attacker/hacker to spy on the user's activities. The attacker can use the machine as part of a botnet and send spam e-mail, distribute pornography, launch distributed denial-of-service attacks, install third-party malware, download/upload files on the machine, log keystrokes, modify or delete files, steal important data, crash the machine, etc.

Hackers use port scanner to find compromised machines on the network and, once found, they install malicious program on such compromised machines to use these networked machines as botnets. Due to the popularity of botnets among hackers, Trojan horse malware is on the rise, accounting for the largest percentage of the global malware detected in the computer network world.

Backdoors

Backdoors in a computer system are remote administration programs that allow hackers to access and control the user's computer while attempting to remain undetected. Common examples of backdoors are BackOrifice, Netbus and SubSeven. Some backdoors rewrite the compiler and piggy-back themselves during the compilation process. Such compromised compiler includes backdoors in the compiled output, keeping the source code of a program intact.

Phishing attacks

Phishing attack is a way to trick computer users into divulging personal authentication data such as username, password and credit card number through a fraudulent e-mail message



or website. These attacks involve mass distribution of spoofed e-mail messages that have links to the fraudulent websites and seem to come from a trusted source, such as a bank, big reputed merchant and trusted service provider. Upon access by the recipient, these fraudulent websites ask to provide personal information, which is later used for identity theft.

Cross-site scripting attacks

Security of the Web is somewhat based on the same origin policy that states that "if it is learnt that content from a trusted website is granted permission to access resources on a user's system, then any content from that website will share the same resources, while content from some other website will have to get permission separately to access the resources."

Hackers/attackers make use of some known vulnerability—like cross-site scripting—in a Web-based application to bypass access controls such as same origin policy and launch attacks by injecting client-side script into webpages viewed by other users. Thus, by cross-site scripting, an attacker gains elevated access privileges to sensitive page contents, session cookies, etc.

Attacks associated with cookies

Cookies are small text files that are created when the user visits a website. The website uses the cookie to

remember who the user is and what activity has been performed by the user recently. The website retrieves the cookie from the hard disk the next time the user visits the same website. In this way, cookies are very helpful as the user doesn't have to type the same information every time.

However, vulnerability in the browser may allow hackers to access important information stored in these cookies that hackers may use to gain advantage.

Invasive software

All kinds of software that arrive on a user's computer without permission or awareness of the user, or sometimes with user permission, are known as invasive software. Typical forms of invasive software are adware, spyware, scareware, scumware, theftware and drug dealerware. These software may pollute the user's screen with ads and popups, send user information, slow down the user's computer or even cause system crash.

Adware. Adware is designed to advertise a commercial offering. It is commonly acquired when a user downloads freeware or shareware like games. When the user downloads the software, the user has to click through a page of unintelligible legal stuff that includes some kind of copyright and usually permission to install adware along with the software. After downloading the software, the user may start noticing more pop-up windows than usual.

Spyware. Spyware is a software that is more likely engaged in antisocial activities such as sending personal information like passwords, credit card numbers and other confidential corporate information to its creators. Spyware installs itself without the user's permission and often hides so that it is difficult to find and eradicate it from the system.

One of the threats of spyware is